

Karthik Bhattaram

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EDUCATION

University of California, Santa Barbara

Santa Barbara

Master of Science in Computer Science

March 2026 to March 2027

- Courses: Foundation Models, Classic Papers in Security, Frontier LLM and Agent Capabilities
- Researching machine learning and cybersecurity under Christopher Kruegel and Giovanni Vigna

University of California, Santa Barbara

Santa Barbara

Bachelor of Science in Computer Science

September 2022 to March 2026

- Courses: Deep Learning, Machine Learning for Networking, Artificial Intelligence
- Honors: Regents Scholar, College of Engineering Honors, Distinction in Major

EXPERIENCE

Security Research Assistant

October 2025 – June 2026

Shellphish

Santa Barbara, California

- Built a benchmark of 8 Intrusion Detection Systems against 9,700+ C2 beacon samples spanning 10 frameworks
- Showed learned behavioral detectors achieve near perfect incident recall while signature-based methods fall short

Software Engineer Intern

June 2025 – September 2025

Arista Networks

Santa Clara, California

- Accelerated Broadcom switching ASIC diagnostics by orders of magnitude using custom C++ utilities
- Developed an interprocess communication tool for Microsoft and OpenAI datacenters

Machine Learning Research Intern

September 2024 – June 2025

University of California, Santa Barbara

Santa Barbara, California

- Published a paper titled “Leveraging Large Language Models to Contextualize Network Measurements”
- Aggregated FCC data into a SQL database to improve model context using retrieval-augmented generation

Software Engineer Intern

July 2024 – September 2024

Cadence Design Systems / OpenEye Scientific

Santa Fe, New Mexico

- Wrote AWS-based package to benchmark database compression algorithms that saved terabytes of server space
- Updated CUDA C++ software to incorporate new language features to match competing industry solutions

Hardware Architecture Intern

June 2023 – September 2023

Cadence Design Systems

San Jose, California

- Implemented a Python automation tool to generate memory wrappers for use on a hardware emulation platform
- Collected foundry provided memories written in System Verilog to be used as templates for this tool

Security Software Intern

June 2021 – August 2022

Axiado Corporation

San Jose, California

- Developed a Python package to create network packet flows for use in hardware simulation and emulation
- Created and executed a test plan for a hardware firewall’s parser module and discovered multiple bugs

PROJECTS

Graph Generation Fine Tuning | *Python, PyTorch, Graph Algorithms*

January 2026 – March 2026

- Adapted Meta’s Llama 3.1 model to process graph generation tasks using parameter-efficient fine tuning
- Compared zero-shot, supervised fine tuning, direct preference optimization, and steering performance
- Achieved 85% improvement in graph correctness (based on graph metrics) over the zero shot model

TECHNICAL SKILLS

Certificates: Enterprise Security Professional, Python for Data Science and AI

Languages: Python, C/C++, Java, JavaScript, Shell, Verilog

Developer Tools: Git, Visual Studio Code, Vim, Emacs

Libraries: Pandas, NumPy, PyTest, Black, Matplotlib